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STAT525 HOMEWORK#4

1. Consider the following data set that describes the relationship between "velocity" of an enzymatic reaction (V) and the substrate concentration (C).

Concentration	Velocity
0.02	76
0.02	47
0.06	97
0.06	107
0.11	123
0.11	139
0.22	159
0.22	152
0.56	191
0.56	201
1.10	207
1.10	200

A common model used to describe the relationship between "velocity" and concentration is the Michaelis-Menten model

$$V = \frac{\theta_1 C}{\theta_2 + C}$$

where θ_1 is the maximum velocity of the reaction and θ_2 describes how quickly (in terms of increasing concentration) the reaction will reach maximum velocity. With this model, $1/V$ can be written as a linear model with explanatory variable $1/C$.

$$\frac{1}{V} = \frac{1}{\theta_1} + \frac{\theta_2}{\theta_1}\left(\frac{1}{C}\right) = \beta_0 + \beta_1\left(\frac{1}{C}\right)$$

You are asked to investigate whether this linear transformation results in a good or poor fit by doing the following steps:

a. Generate a scatterplot of V vs C. Comment on the shape.

b. Define new variables for $1/V$ and $1/C$ in SAS and generate a scatterplot. These new variables can be defined as follows:

```
DATA NEW;
    INPUT c v;
    vinv = 1/v;
    cinv = 1/c;
    CARDS;
    [PUT DATA HERE]
;
```

Does the fit appear linear? Does there appear to be any violation of assumptions?

c. Is the distribution of $1/C$ different than C? Are there any points that may be more influential in determining the fit?

d. Determine the least squares regression line for $1/V$ vs $1/C$. Save the residuals and predicted values. Does the residual plot suggest any problems?

e. Convert this regression line back into the original nonlinear model and plot the predicted curve on a scatterplot of V vs C. Comment on the fit. To generate the predicted curve, simply take the predicted values from the regression model and "re-inverse" them. For example, consider new1 to be the data set containing the residuals and predicted values.

```
DATA new1;
    SET new1;
    pred1 = 1/pred;

SYMBOL1 V=CIRCLE I=NONE C=BLACK;
SYMBOL2 V=PLUS I=SM5 C=BLACK;
PROC GPLOT;
    PLOT v*c=1 pred1*c=2 / OVERLAY;
RUN;
```

2. KNNL Problem 4.12. This can be done in SAS or by hand. To fit a model with no intercept in SAS, you need to add the `NOINT` option on `MODEL` statement line.

3. KNNL Problem 4.21

4. KNNL Problem 4.25 (STAT Majors)

5. KNNL Problem 6.2

6. (SAS Exercise) Use the **brand preference** data described in KNNL Problem 6.5. Run the linear regression with moisture and sweetness of the product as the explanatory variables and degree of liking as the response variable.

   a. Summarize the regression results by giving the fitted regression equation, the value of R^2.

   b. State the results of the significance test for the null hypothesis that the two regression coefficients for the explanatory variables are *all* zero (give null and alternative hypotheses, test statistic with degrees of freedom, p-value, and a brief conclusion in words).

   c. Describe the results of the hypothesis tests for the individual regression coefficients (give null and alternative hypotheses, test statistic with degrees of freedom, p-value, and a brief conclusion in words).

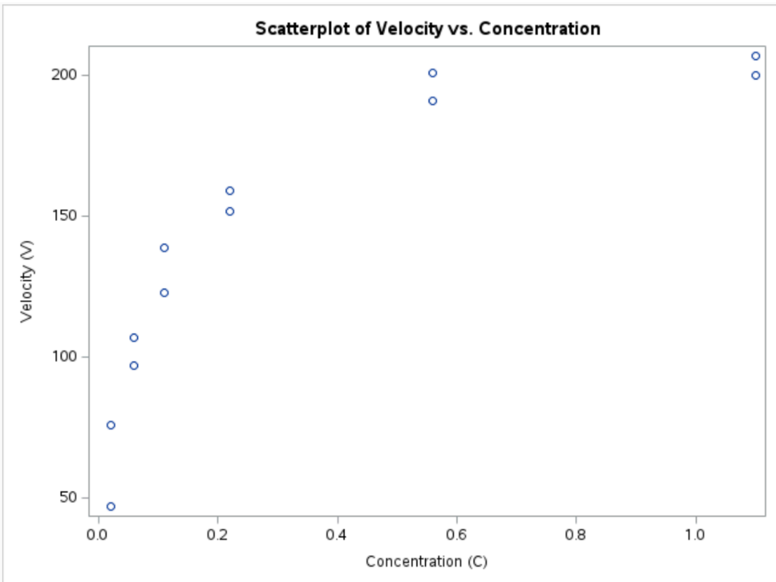
   d. Give separate 95% confidence intervals for the regression coefficients of sweetness and moisture. What is the relationship between these confidence intervals and the above hypothesis results?

1. Consider the following data set that describes the relationship between "velocity" of an enzymatic reaction (V) and the substrate concentration (C).

```
data enzyme;
    input concentration velocity;
    datalines;
0.02 76
0.02 47
0.06 97
0.06 107
0.11 123
0.11 139
0.22 159
0.22 152
0.56 191
0.56 201
1.10 207
1.10 200
;
run;
```

a. Generate a scatterplot of V vs C. Comment on the shape.

```
proc sgplot data=enzyme;
    scatter x=concentration y=velocity;
    xaxis label="Concentration (C)";
    yaxis label="Velocity (V)";
    title "Scatterplot of Velocity vs. Concentration";
run;
```



The relationship is nonlinear:

- ↑ rapidly at low concentrations and begins to plateau as concentration ↑, hence consistent with the Michaelis-Menten kinetics.

- Michaelis-Menten kinetics: reaction velocity ↑ quickly at first but approaches a maximum velocity (θ_1) as substrate concentration ↑.

- Shape is concave down (decelerating increase), ↑C, ↓ in gains of V.

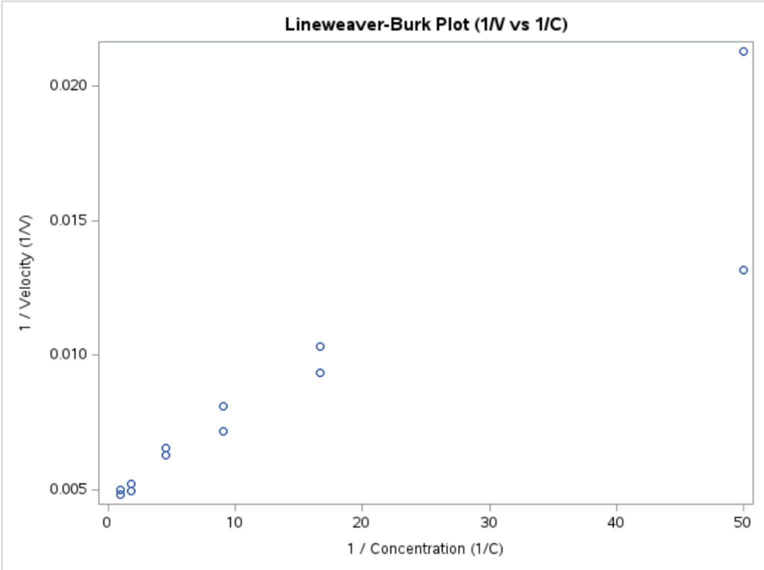
- Linear model would not be appropriate, should opt for nonlinear / transformed model.

b. Define new variables for $1/V$ and $1/C$ in SAS and generate a scatterplot. These new variables can be defined as follows:

Does the fit appear linear? Does there appear to be any violation of assumptions?

```
data enzyme2;
    set enzyme;
    vinv = 1 / velocity;
    cinv = 1 / concentration;
run;

proc sgplot data=enzyme2;
    scatter x=cinv y=vinv;
    xaxis label="1 / Concentration (1/C)";
    yaxis label="1 / Velocity (1/V)";
    title "Lineweaver-Burk Plot (1/V vs 1/C)";
run;
```



Yes. Transformed data appear to form a rough linear trend, especially for mid to high concentrations.

• Violations?:

◦ There are 2 points on the far right that appear far from the cluster, coming from the lowest concentration, C, but largest 1/C.

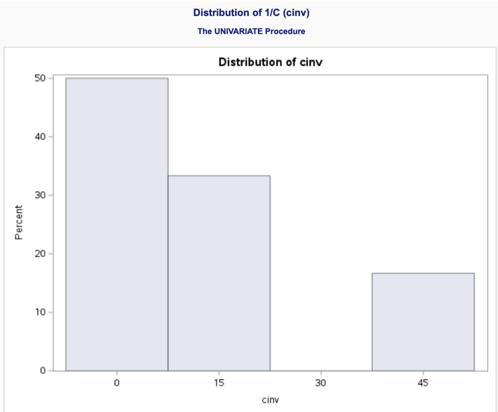
• These points can exert ↑ leverage and may influence the regression line disproportionally.

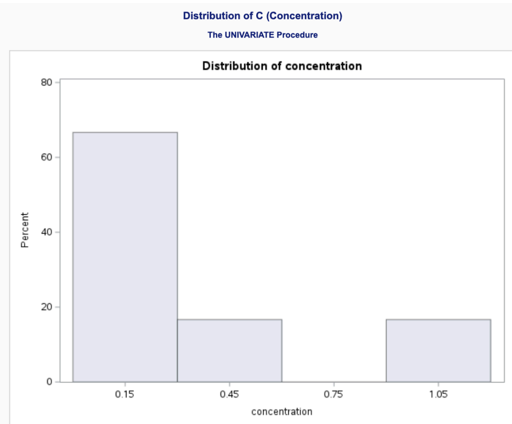
• Suggesting a possible violation of equal variance (heteroscedasticity) and potential influential outliers.
↓
1/C=50

c. Is the distribution of $1/C$ different than C? Are there any points that may be more influential in determining the fit?

```
proc univariate data=enzyme;
    var concentration;
    histogram;
    qqplot;
    title "Distribution of C (Concentration)";
run;

proc univariate data=enzyme2;
    var cinv;
    histogram;
    qqplot;
    title "Distribution of 1/C (cinv)";
run;
```





Yes. Distribution for 1/C is much more skewed and stretched than the Distribution for C.

◦ The smallest C (especially C=0.02) result in very large 1/C, which may act as influencial points and could potentially distort the linear regression fit.

d. Determine the least squares regression line for $1/V$ vs $1/C$. Save the residuals and predicted values. Does the residual plot suggest any problems?

```
proc reg data=enzyme2 outest=est;
    model vinv = cinv;
    output out=reg_out p=vinv_pred r=resid;
    title "Regression: 1/V vs 1/C";
run;
quit;

/* Residual plot */
proc sgplot data=reg_out;
    scatter x=cinv y=resid;
    refline 0 / axis=y lineattrs=(color=red);
    xaxis label="1 / Concentration (1/C)";
    yaxis label="Residuals";
    title "Residual Plot from Lineweaver-Burk Regression";
run;
```

Regression: 1/V vs 1/C

The REG Procedure
Model: MODEL1
Dependent Variable: vinv

Number of Observations Read	12
Number of Observations Used	12

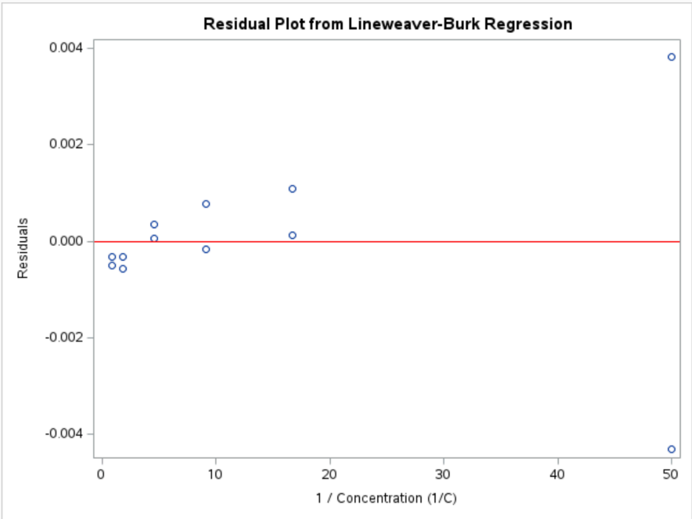
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	0.00021232	0.00021232	59.30	<.0001
Error	10	0.00003581	0.00000358		
Corrected Total	11	0.00024813			

Root MSE	0.00189	R-Square	0.8557
Dependent Mean	0.00853	Adj R-Sq	0.8413
Coeff Var	22.19144		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	0.00511	0.00070400	7.25	<.0001
cinv	1	0.00024722	0.00003210	7.70	<.0001



Fitted Regression Line:

$$\widehat{1/V} = 0.00511 + 0.00024722 \cdot (1/C)$$

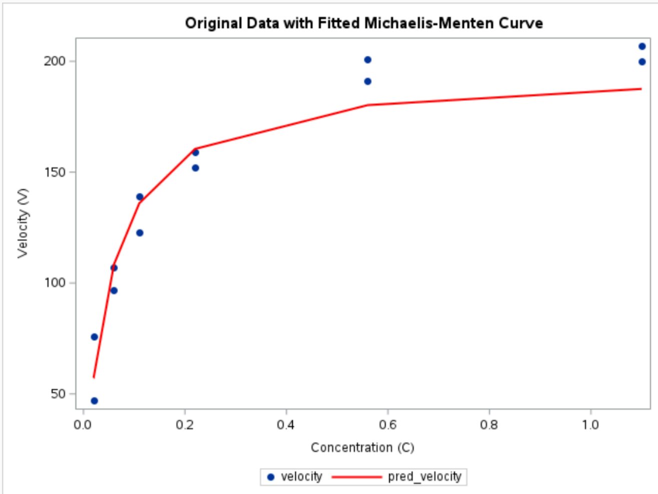
problem!
- Points do not scatter randomly around line at 0.
- e^2 ↑ in conical fashion.

- Model explains a large proportion of the variation in $1/V$ ($R^2 = 0.8577$)
- e^2 plot shows 2 extreme values with large residuals, likely due to influence from very small C's (e.g. $C = 0.02$).
- These extreme values may impact the stability of the model and violate equal variance (heteroscedasticity).

e. Convert this regression line back into the original nonlinear model and plot the predicted curve on a scatterplot of V vs C. Comment on the fit. To generate the predicted curve, simply take the predicted values from the regression model and "re-inverse" them. For example, consider new1 to be the data set containing the residuals and predicted values.

```
data reg_out2;
    set reg_out;
    pred_velocity = 1 / vinv_pred;
run;

proc sgplot data=reg_out2;
    scatter x=concentration y=velocity / markerattrs=(symbol=circlefilled);
    series x=concentration y=pred_velocity / lineattrs=(color=red thickness=2);
    xaxis label="Concentration (C)";
    yaxis label="Velocity (V)";
    title "Original Data with Fitted Michaelis-Menten Curve";
run;
```



- Fitted curve from the Michaelis-Menten model aligns well with the original V, especially at mid-to-high C's.
- The shape of the fitted line captures the expected saturation effect of enzyme kinetics.
- Minor deviations are seen at ↓C's, but overall the model provides a good fit and represent the underlying relationship well.

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2. KNNL Problem 4.12. This can be done in SAS or by hand. To fit a model with no intercept in SAS, you need to add the `NOINT` option on `MODEL` statement line.

4.12. Typographical errors. Shown below are the number of galleys for a manuscript (X) and the total dollar cost of correcting typographical errors (Y) in a random sample of recent orders handled by a firm specializing in technical manuscripts. Since Y involves variable costs only, an analyst wished to determine whether regression-through-the-origin model (4.10) is appropriate for studying the relation between the two variables.

i:	1	2	3	4	5	6	7	8	9	10	11	12
X_i:	7	12	10	10	14	25	30	25	18	10	4	6
Y_i:	128	213	191	178	250	446	540	457	324	177	75	107

a. Fit regression model (4.10) and state the estimated regression function.

```
data typos;
    input x y;
    datalines;
7   128
12  213
10  191
10  178
14  250
25  446
30  540
25  457
18  324
10  177
4   75
6   107
;
run;

/* Fit regression through the origin */
proc reg data=typos;
    model y = x / noint;
    title "Regression Through the Origin: Typo Cost vs. Galleys";
run;
```

Regression Through the Origin: Typo Cost vs. Galleys

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	12
Number of Observations Used	12

Note: No intercept in model. R-Square is redefined.

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1044939	1044939	51446.2	<.0001
Error	11	223.42426	20.31130		
Uncorrected Total	12	1045162			

Root MSE	4.50681	R-Square	0.9998
Dependent Mean	257.16667	Adj R-Sq	0.9998
Coeff Var	1.75248		

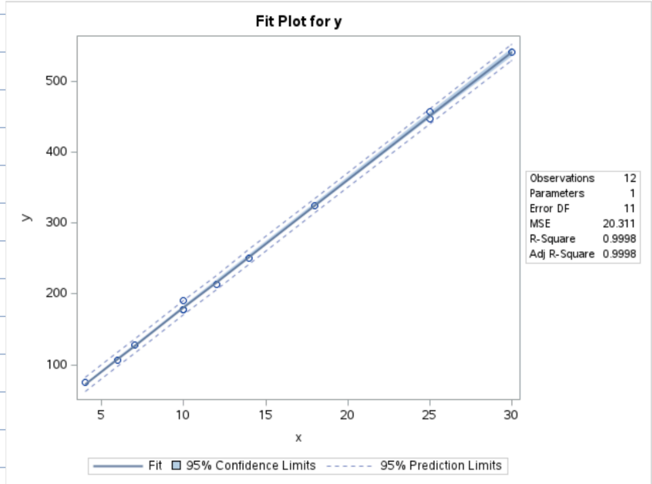
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
x	1	18.02830	0.07948	226.82	<.0001

Fitted Regression Line: $\hat{Y} = 18.02830 \cdot X$

b. Plot the estimated regression function and the data. Does a linear regression function through the origin appear to provide a good fit here? Comment.

```
/* (2b) Scatterplot with fitted regression line through origin */
proc sgplot data=typos;
    scatter x=x y=y / markerattrs=(symbol=circlefilled);
    reg x=x y=y / noint lineattrs=(color=red thickness=2);
    xaxis label="Number of Galleys (X)";
    yaxis label="Typo Correction Cost (Y)";
    title "Regression Through the Origin: Cost vs. Galleys";
run;
```



- The fit plot shows the regression-through-the-origin model provides an excellent fit to the data.
- Observed points align closely with fitted line, with very narrow CI and PI.
- The ↑↑↑ R^2 (0.9998), with visual evidence, confirms linear model without intercept is highly appropriate for modeling # of galley v.s. typo $.

c. In estimating costs of handling prospective orders, management has used a standard of \$17.50 per galley for the cost of correcting typographical errors. Test whether or not this standard should be revised; use $\alpha = .02$. State the alternatives, decision rule, and conclusion.

$H_0: \beta_1 = 17.50$ (standard)
$H_A: \beta_1 \neq 17.50$ (non-standard)

$\boxed{t^* = \frac{\hat{\beta}_1 - H_0}{SE}} = \frac{18.02830 - 17.50}{0.07948} = 6.64696 >>> 2.718$ $\quad \boxed{t_{\alpha/2, df}} = t_{0.01, 11} = 2.718$

- Reject H_0. At $\alpha = 0.02$, $t^* > t$
- There is sufficient evidence to conclude true cost / gallery ≠ \$17.50, so standard should be revised.

d. Obtain a prediction interval for the correction cost on a forthcoming job involving 10 galleys. Use a confidence coefficient of 98 percent.

```
data predict;
    input x;
    datalines;
10
;
run;

data all;
    set typos predict;
run;

proc reg data=all;
    model y = x / noint clm cli alpha=0.02; /* 98% confidence (a
    output out=pred_out p=predicted lclm=lcl_mean uclm=ucl_mean
                   lcli=lcl_pred ucli=ucl_pred;
    title "98% Prediction Interval for X = 10 Galleys";
run;
```

98% PI for X=10: [167.84, 192.72]

98% Prediction Interval for X = 10 Galleys

The REG Procedure
Model: MODEL1
Dependent Variable: y

Output Statistics

Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	98% CL Mean		98% CL Predict		Residual
1	128	126.1981	0.5564	124.6858	127.7104	113.8553	138.5410	1.8019
2	213	216.3397	0.9538	213.7471	218.9322	203.8185	228.8608	-3.3397
3	191	180.2830	0.7948	178.1226	182.4435	167.8441	192.7220	10.7170
4	178	180.2830	0.7948	178.1226	182.4435	167.8441	192.7220	-2.2830
5	250	252.3963	1.1128	249.3717	255.4209	239.7785	265.0140	-2.3963
6	446	450.7076	1.9871	445.3065	456.1087	437.3199	464.0953	-4.7076
7	540	540.8491	2.3845	534.3679	547.3304	526.9904	554.7079	-0.8491
8	457	450.7076	1.9871	445.3065	456.1087	437.3199	464.0953	6.2924
9	324	324.5095	1.4307	320.6207	328.3983	311.6572	337.3618	-0.5095
10	177	180.2830	0.7948	178.1226	182.4435	167.8441	192.7220	-3.2830
11	75	72.1132	0.3179	71.2490	72.9774	59.8329	84.3935	2.8868
12	107	108.1698	0.4769	106.8736	109.4661	95.8516	120.4881	-1.1698
13	.	180.2830	0.7948	178.1226	182.4435	167.8441	192.7220	.

3. KNNL Problem 4.21

4.21. When the predictor variable is so coded that $\bar{X} = 0$ and the normal error regression model (2.1) applies, are b_0 and b_1 independent? Are the joint confidence intervals for β_0 and β_1 then independent?

Yes. $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent when $\bar{x} = 0$, and normal assumption holds.

- Under standard simple linear regression model, the covariance $\hat{\beta}_0$ and $\hat{\beta}_1$: $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-\bar{x} \cdot \sigma^2}{S_{xx}}$
  so if $\bar{x} = 0$, $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0$.

However, the joint CI for $\hat{\beta}_0, \hat{\beta}_1$ are not independent.

- A joint CI accounts for simultaneous variability, and doesn't break into 2 independent intervals, even if $\hat{\beta}_0$ and $\hat{\beta}_1$ are uncorrelated.
  ↓
  (elliptical region under multivariate t-distribution)

4. KNNL Problem 4.25 (STAT Majors) (Skipped)

4.25. Derive the formula for $s^2\{\hat{Y}_h\}$ given in Table 4.1 for linear regression through the origin.

Model:

We assume the linear regression **through the origin**:

$$Y_i = \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim \text{iid } N(0, \sigma^2)$$

Let:

- $\hat{\beta}_1 = \frac{\sum X_i Y_i}{\sum X_i^2}$
- $\hat{Y}_h = \hat{\beta}_1 X_h$ be the predicted value at point X_h

Final Answer:

$$\boxed{s^2\left\{\hat{Y}_h\right\} = s^2 \cdot \frac{X_h^2}{\sum X_i^2}}$$

Step 1: Variance of the prediction

$$\mathrm{Var}(\hat{Y}_h) = \mathrm{Var}(\hat{\beta}_1 X_h) = X_h^2 \cdot \mathrm{Var}(\hat{\beta}_1)$$

Step 2: Variance of $\hat{\beta}_1$ in regression through the origin:

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum X_i^2}$$

So:

$$\mathrm{Var}(\hat{Y}_h) = X_h^2 \cdot \frac{\sigma^2}{\sum X_i^2}$$

Step 3: Estimate σ^2 using the MSE:

$$s^2 = \frac{\text{SSE}}{n-1}$$

5. KNNL Problem 6.2

6.2. Set up the $\mathbf{X}$ matrix and $\boldsymbol{\beta}$ vector for each of the following regression models (assume $i = 1, \ldots, 5$):

a. $Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1}^2 + \varepsilon_i$

b. $\sqrt{Y_i} = \beta_0 + \beta_1 X_{i1} + \beta_2 \log_{10} X_{i2} + \varepsilon_i$

a) $Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1}^2 + \varepsilon_i$

$$X = \begin{bmatrix} X_{11} & X_{12} & X_{11}^2 \\ X_{21} & X_{22} & X_{21}^2 \\ X_{31} & X_{32} & X_{31}^2 \\ X_{41} & X_{42} & X_{41}^2 \\ X_{51} & X_{52} & X_{51}^2 \end{bmatrix} \quad \beta_1 = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

b) $\sqrt{Y_i} = \beta_0 + \beta_1 X_{i1} + \beta_2 \log_{10} X_{i2} + \varepsilon_i$

$$X = \begin{bmatrix} 1 & X_{11} & \log_{10}(X_{12}) \\ 1 & X_{21} & \log_{10}(X_{22}) \\ 1 & X_{31} & \log_{10}(X_{32}) \\ 1 & X_{41} & \log_{10}(X_{42}) \\ 1 & X_{51} & \log_{10}(X_{52}) \end{bmatrix} \quad \beta_1 = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} \quad Y = \begin{bmatrix} \sqrt{Y_1} \\ \sqrt{Y_2} \\ \sqrt{Y_3} \\ \sqrt{Y_4} \\ \sqrt{Y_5} \end{bmatrix}$$

6. (SAS Exercise) Use the **brand preference** data described in KNNL Problem 6.5. Run the linear regression with moisture and sweetness of the product as the explanatory variables and degree of liking as the response variable.

6.5. Brand preference. In a small-scale experimental study of the relation between degree of brand liking (Y) and moisture content (X_1) and sweetness (X_2) of the product, the following results were obtained from the experiment based on a completely randomized design (data are coded):

i:	1	2	3	...	14	15	16
X_{i1}:	4	4	4	...	10	10	10
X_{i2}:	2	4	2	..	4	2	4
Y_i:	64	73	61	...	95	94	100

a. Summarize the regression results by giving the fitted regression equation, the value of R^2.

```
data brand_pref;
    input liking moisture sweetness;
    datalines;
64  4 2
73  4 4
61  4 2
76  4 4
72  6 2
80  6 4
71  6 2
83  6 4
83  8 2
89  8 4
86  8 2
93  8 4
88 10 2
95 10 4
94 10 2
100 10 4
;
run;

proc reg data=brand_pref;
    model liking = moisture sweetness;
    title "Regression of Liking on Moisture and Sweetness";
run;
```

Regression of Liking on Moisture and Sweetness

The REG Procedure
Model: MODEL1
Dependent Variable: liking

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1872.70000	936.35000	129.08	<.0001
Error	13	94.30000	7.25385		
Corrected Total	15	1967.00000			

Root MSE	2.69330	R-Square	0.9521
Dependent Mean	81.75000	Adj R-Sq	0.9447
Coeff Var	3.29455		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	37.65000	2.99610	12.57	<.0001
moisture	1	4.42500	0.30112	14.70	<.0001
sweetness	1	4.37500	0.67332	6.50	<.0001

Fitted Regression Line: $\widehat{\text{Liking}} = 37.6500 + 4.42500 \cdot \text{Moisture} + 4.37500 \cdot \text{Sweetness}$

$R^2 = 0.9521$

- 95% of variation in degree of liking is explained by Moisture and Sweetness.

b. State the results of the significance test for the null hypothesis that the two regression coefficients for the explanatory variables are *all* zero (give null and alternative hypotheses, test statistic with degrees of freedom, p-value, and a brief conclusion in words).

$H_0: \beta_1 = \beta_2 = 0$

H_A: At least one β_1 or $\beta_2 \neq 0$

(Results from 6a):

- $F = 129.08$
- df1 (model) = 2
- df2 (error) = 13
- p-value: < 0.0001

→ Reject H_0 at any α level.

- There is strong evidence that at least one β (Moisture or Sweetness) is significantly associated with degree of liking.

c. Describe the results of the hypothesis tests for the individual regression coefficients (give null and alternative hypotheses, test statistic with degrees of freedom, p-value, and a brief conclusion in words).

Moisture

$H_0: \beta_1 = 0$
$H_A: \beta_1 \neq 0$

- $t^* = \frac{4.4250}{0.30112} = 14.70$
- df = 13
- p-value: $p < 0.0001$

Reject H_0. Moisture is significantly associated with degree of liking.

Sweetness

$H_0: \beta_2 = 0$
$H_A: \beta_2 \neq 0$

- $t^* = \frac{4.3750}{0.67332} = 6.50$
- df = 13
- p-value: $p < 0.0001$

Reject H_0. Sweetness is significantly associated with degree of liking.

Both Moisture or Sweetness have strong evidence of significantly associated with degree of liking.

d. Give separate 95% confidence intervals for the regression coefficients of sweetness and moisture. What is the relationship between these confidence intervals and the above hypothesis results?

$t_{0.025,13} = 2.160$

Moisture

(Results from 6a):
- Estimate: 4.4250
- SE: 0.30112
- df: 13

CI: $4.4250 \pm 2.160 \times 0.30112 = 4.4250 \pm 0.6504$

$\Downarrow$

$[3.7746, 5.0754]$ Moisture

Sweetness

(Results from 6a):
- Estimate: 4.3750
- SE: 0.67332
- df: 13

CI: $4.3750 \pm 2.160 \times 0.67332 = 4.3750 \pm 1.4544$

$\Downarrow$

$[2.9206, 5.8294]$ Sweetness

- Both CI do not contain 0, matching conclusion in 6 (c). $\rightarrow$ Rejected H_0: All $\beta = 0$.